Portable Arduino Based Single Lead ECG and Heartbeat Monitoring System

Amir Haffar, Nasser Trad, Daniel Rushakoff

***Abstract*—A compact and low cost heart monitoring system is presented that can capture single lead electrocardiogram (ECG) signals and measure heart rate in real time using an Arduino-based platform. The proposed system integrates an Arduino UNO R4 WiFi microcontroller with a MAX30102 pulse oximeter sensor and a commercial ECG module to collect blood oxygen saturation (SpO_2), ECG waveforms, and heart rate data in real time. Basic filtering and peak detection algorithms are implemented to reduce noise and compute the heartbeat rate from the sensor data. The processed signals and calculated BPM are visualized live using the Arduino Serial Plotter and an IoT cloud dashboard. We demonstrate that the device, built with approximately $50 of components, can successfully capture key features of the ECG waveform and track heart rate with an estimated accuracy of about 80% to 85% compared to medical grade equipment which we know has higher rate of accuracy. This portable monitoring system illustrates the feasibility of accessible, real-time heart health tracking, with potential for future enhancements in signal processing and wireless connectivity.**

***Index Terms*—Biomedical instrumentation, ECG, heart rate, Arduino, AD8232, MAX30102, IoT, wearable health.**

I. Introduction

Cardiovascular diseases remain the leading cause of death globally, responsible for approximately 19.8 million deaths in 2022 according to the World Health Organization (WHO). Early detection of cardiac abnormalities is crucial for timely intervention, especially in lower income communities that lack access to advanced diagnostic and health monitoring tools. Electrocardiography (ECG) and heart rate monitoring are primary diagnostic tools to assess cardiac function. The ECG waveform reflects the heart's electrical activity, with the P wave representing atrial depolarization, the QRS complex corresponding to ventricular depolarization, and the T wave indicating ventricular repolarization.

These features, shown in Figure 1, provide insights into heart rhythm and can reveal arrhythmias, ischemia, and other abnormalities.

In clinical settings, multi-lead ECG machines provide heart monitoring with high accuracy. However, such equipment is expensive and not accessible outside hospitals. There is a growing demand for low cost, devices that can track the user or patients health in everyday settings. The popularity of consumer wearables such as smartwatches and fitness band trackers has increased significantly over the past few years, especially after the COVID-19 pandemic in 2020. Several research efforts have explored IoT based ECG monitoring systems to enable remote cardiac monitoring [5].

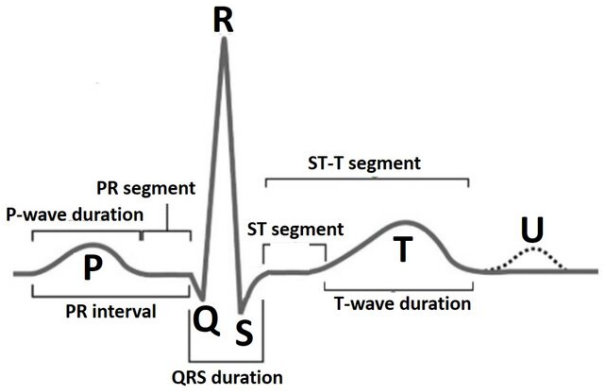


Fig. 1. ECG waveform showing the P wave, QRS complex, T wave, and U wave, along with corresponding durations and segments such as the PR interval, ST segment, QRS duration. These provide an insight into heart rhythm and electrical conduction, which are critical for detecting arrhythmias, and other abnormalities.

This paper presents a portable single-lead ECG and heartbeat monitoring system. The system is built around an Arduino UNO R4 WiFi microcontroller and integrates two sensors the MAX30102 optical heart rate/SpO_2 sensor and the AD8232 single lead ECG analog front end. The goal is to create an affordable, compact device that can capture real time heart signals by recording the heart rate waveform and simultaneously reporting the user's BPM and SpO_2,bridging the gap between clinical monitors and personal wearable devices. In the following sections, we will discuss the problem, design requirements, our technical approach, implementation results, and conclusions with future improvements.

II. Problem Description

Continuous heart monitoring is essential for early detection of cardiac issues, but traditional 12 lead ECG setups are impractical for everyday use. The problem addressed by this project is the development of a simpler version using a single lead ECG and heart rate monitor that is portable and inexpensive, yet provides useful cardiac information.A single lead ECG uses three surface electrodes, with two placed just below the right and left clavicles and a third on the left side of the rib cage. This configuration can capture basic heart rhythm and rate, but it does not provide the full waveform detail available from a standard 12 lead ECG. By combining a single lead ECG with heart rate and SpO_2 measurements from the MAX30102 pulse oximeter, the system can cross

validate heart rate and provide additional information about blood oxygen saturation.

The main challenges of this project started with amplifying and filtering the small ECG signals from the body to obtain a clear waveform. The second problem was dealing with noise artifacts from muscle movement, power interference, and electrode contact. Our third and final issue was implementing a reliable peak detection algorithm on an embedded platform with limited resources. Additionally, the device needed to display or transmit the data for real time monitoring on the dashboard we created for our wireless interface.

The design objectives we defined for ourselves for the project:

- Create a cost-effective prototype using off the shelf sensors being the MAX30102 and AD8232 as well as using an Arduino microcontroller to measure BPM, single-lead ECG signals and SpO_2.
- Integrate the MAX30102 and AD8232 sensors with the Arduino UNO R4 WiFi on a single platform, ensuring proper wiring, stable power, and a common ground for reliable readings.
- Capture heart rate, SpO_2, and ECG signals in real time and visualize them on a computer via the Arduino Serial Plotter (and optionally on a cloud dashboard).
- Evaluate the system's performance in terms of signal quality and heart rate accuracy, and identify its limitations compared to medical devices.

III. Technical Approach

A. Hardware Design

The hardware consists of an Arduino UNO R4 WiFi board connected to a MAX30102 pulse oximeter/heart-rate sensor and an AD8232 ECG module, along with three ECG electrodes. Fig. 2 shows the prototype wiring of the system on a breadboard and Fig. 3 is our completed circuit with everything connected. The Arduino UNO R4 WiFi was chosen for its compatibility with the sensors and built in wireless connectivity for future expansion.The MAX30102 is a noninvasive optical Photoplethysmography sensor (PPG) that measures pulse at the fingertip and estimates blood oxygen saturation (SpO_2) by using light to detect changes in blood volume in the skin's microvascular tissue bed.

It interfaces with the Arduino over I^2C using the SDA and SCL lines and is powered at 5V. The AD8232 module is a single lead ECG analog sensor that amplifies and filters the electrical signals of the heart to produce a readable waveform. The AD8232 outputs an analog voltage representing the ECG waveform, which we connected to the Arduino's analog input A0. The module can be powered by 3.3 V or 5 V and, in our setup, it was powered by 5 V and all components shared a common ground reference. In addition to powering the board over USB during development, we were also able to run the system from a 9 V battery, which allowed us to use it wirelessly and stream data to the Arduino cloud dashboard without keeping it physically connected to a computer.

For the ECG electrodes, we used three adhesive pads in a standard setup placing two on the upper chest near the clavicles for our differential measurement and one on the left side of the abdomen as a reference. This setup is the standard single lead ECG configuration. The three electrode leads from the AD8232 module connect directly to these pads. The AD8232 chip includes instrumentation amplifiers and filter circuits such as a high pass filter to remove any baseline drift and low pass filter to help limit noise to clean up the ECG signal before it reaches the Arduino's ADC. The module also has an LED that flashes with each detected heartbeat, which is helpful for debugging.

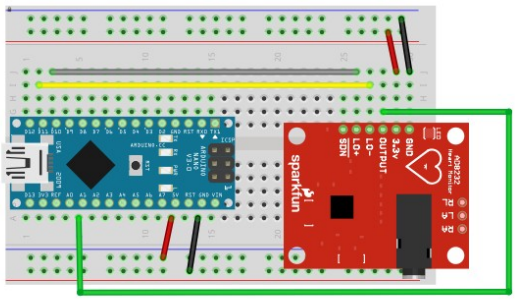


Fig. 2. Breadboard prototype of the heart monitoring system showing the Arduino UNO R4 WiFi (blue board) connected to the AD8232 ECG module (red board). The wiring includes power 5V and GND and the analog ECG output from the AD8232 into Arduino A0, with all components sharing a common ground.

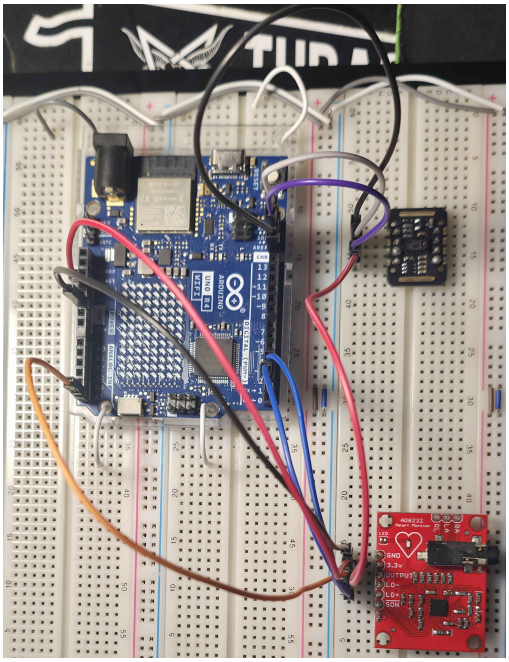


Fig. 3. Hardware prototype of the heart monitoring system. The Arduino UNO R4 WiFi is mounted on the breadboard and wired to the AD8232 ECG module (red board) and the MAX30102 pulse oximeter sensor (small board at the top right).

B. Software and Signal Processing

The microcontroller software was developed using the Arduino IDE and C++ code and we implemented a single file that initializes both the MAX30102 pulse oximeter and the AD8232 ECG module. The Arduino initializes the MAX30102 sensor using a library for heart rate and SpO_2 measurement and configures the ADC for reading the AD8232 output. The sampling rate for the ECG analog signal was on the order of a few hundred Hz, sufficient to capture the essential shape of the ECG waveform which has most energy below 40Hz. The MAX30102 sensor was configured to sample the PPG at around 50Hz to 100Hz and compute instantaneous heart rate using its internal algorithm or by peak detection on the IR signal.

In the loop, the Arduino first reads the infrared signal from the MAX30102 is roughly 50Hz and using a 20ms delay and passes each sample to the `checkForBeat()` helper function. When a beat is detected, the time difference to the previous beat is measured and converted into an instantaneous heart rate (BPM $= 60/\Delta t$). To prevent the BPM from jumping around, we store a small number of recent beats in an array and display the average of those values as the current heart rate. At the same time, IR and red LED samples are stored in small buffers, and whenever the buffer fills, the Maxim SpO_2 algorithm is called to estimate oxygen saturation and mark whether the reading is valid.

A simplified section of the combined main loop is shown below:

```
// MAX30102 IR
long irValue = particleSensor.getIR();
// AD8232 ECG
long ecgValue = analogRead(ECG_PIN);

if (checkForBeat(irValue)) {
  long delta = millis() - lastBeat;
  lastBeat = millis();

// BPM from IR
  float bpm = 60.0 * 1000.0 / delta;
  if (bpm > 20 && bpm < 255) {
    rates[rateSpot++] = (byte)bpm;
    rateSpot %= RATE_SIZE;
  }
}

// ECG detection
if (ecgValue > ecgThreshold &&
millis() - lastEcgBeat > 300) {
  lastEcgBeat = millis();
}
```

A simple digital filtering step is applied to the raw signals to reduce any noise from our collected signals. In our prototype, this was a basic moving average filter over a short window for the PPG signal and a slight smoothing of the ECG signal since the AD8232 already provides a form of analog filtering. This helped us to stabilize the readings by removing any random spikes we got due to any noise from the users body as well as any outside noise from the environment especially since the MAX30102 was very sensitive to finger placement, pressure, and small movements, so even slight changes often caused large fluctuations in the PPG waveform and SpO_2 estimate, making it difficult to obtain consistent data. We did not use any advanced digital filtering like a bandpass filter, for the AD8232. Instead, we mostly relied on the user staying as still as possible, because any movement of the cables or electrodes could distort the ECG signal.

Heartbeat detection is performed by identifying peaks in the signal. We use a peak detection algorithm on the PPG signal by thresholding. The code continuously updates a running average of the signal and detects a beat when the PPG value exceeds the dynamic threshold relative to the average which also indicates that the users measured pulse's peak. Each detected beat triggers a timestamp recording and the time interval between consecutive beats is used to calculate the instantaneous heart rate in beats per minute (BPM): BPM $= 60/\Delta t$, where Δt is the interval in seconds between beats. The Arduino computes and outputs the current BPM and also a short term average BPM for stability.

The Arduino transmits the data via serial to a connected computer for visualization. We formatted the serial output as comma-separated values containing the IR value, the current BPM, and the estimated SpO_2 seen in Fig. 4. This allowed us to use the Arduino Serial Plotter to view the signals in real time while simultaneously monitoring the numeric heart rate and oxygen saturation values. The Arduino UNO R4 WiFi also has network connectivity, so in our implementation we used Arduino IoT Cloud to send the heart rate data to a dashboard we created.

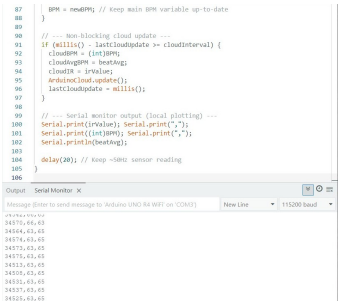


Fig. 4. Serial monitor output showing comma-separated IR, BPM, and SpO_2 values from the MAX30102 sensor.

C. Data Visualization and Cloud Integration

For visualization during testing, we used the Arduino Serial Plotter to view the signals in real time. The ECG waveform from the AD8232 was shown as a live trace, and we could clearly see the QRS peaks in Fig. 5. While tuning the heart rate detection, we mainly watched how the detected BPM and `beatAvg` values responded to changes in the signal and our

threshold settings. Seeing the waveform update live while the BPM changed made it much easier to tell when the detection was working well and when it was missing or double-counting beats, especially when motion or loose electrodes added extra noise.

To enable remote monitoring, we set up an Arduino IoT Cloud dashboard on the Arduino mobile app for our phones. The microcontroller's WiFi capability was used to connect to the Arduino Cloud and update variables for heart rate and, optionally, the raw signals or SpO_2. We built a simple dashboard with widgets for the IR value, instantaneous BPM, average BPM, and a small history panel showing recent heart activity. Fig. 6 shows the mobile version of this dashboard, which lets us monitor the same heart rate data from a phone instead of only through the Serial Plotter on a laptop.



Fig. 5. Arduino Serial Plotter view showing the real-time heartbeat signal, detection threshold, and BPM used for debugging and tuning the peak detection algorithm.

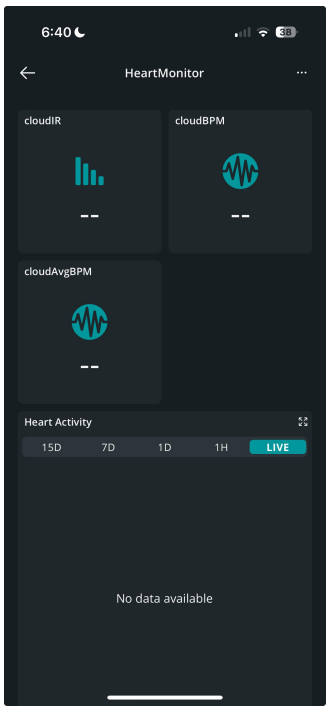


Fig. 6. Arduino IoT Cloud mobile dashboard with widgets for IR (`cloudIR`), instantaneous BPM (`cloudBPM`), and average BPM (`cloudAvgBPM`).

IV. Implementation and Results

The prototype was assembled on a breadboard as previously described and tested on several users sitting at rest in a chair. The system successfully captured a single lead ECG waveform from the AD8232 module. The typical P–QRS–T waveform was visible on the Serial Plotter graph, as shown in Fig. 5, though there was some noise. The R-peaks shows us the ventricular contractions were still clear enough for the peak detection algorithm to register heartbeats. When using the MAX30102 sensor we placed the users finger and it automatically recorded and provided us with the PPG readings from which heart rate was also derived. The real time BPM values from both sensors were in close agreement, usually within a few beats per minute of each other under steady conditions.

One representative run showed a resting heart rate around 65–70 BPM, which is within normal range. The Arduino Cloud dashboard updated the BPM value approximately once per second, confirming the feasibility of remote monitoring through IoT connectivity. The device demonstrated an estimated heart rate accuracy of about 85% compared to a reference pulse oximeter device, which is reasonable given the simple filtering and single lead setup.

However, several challenges and limitations were observed. Because of the minimal filtering in the current design, the ECG signal was susceptible to noise and interference. Even small movements of the subject's arms or torso, or jostling of the electrode wires, introduced large artifacts in the ECG trace. These motion artifacts sometimes obscured the true ECG waveform temporarily. Similarly, the PPG readings from the MAX30102 was to sensitive to how the finger was placed pressing too hard or too lightly on the sensor altered the waveform amplitude and could lead to incorrect heart rate readings.

Because our prototype used only basic software filtering and a simple peak detection, it is not as robust as clinical-grade monitors. For instance, occasional false peaks (due to noise) led to momentary false heart rate spikes. The single-lead ECG provides limited diagnostic information: while it can reliably measure heart rate and reveal gross rhythm abnormalities, it cannot detect certain conditions that require multiple lead views (since it lacks the spatial information of a full 12-lead ECG).

Despite these limitations, the system still showed that we could capture heart signals in real time. We were able to watch the ECG waveform and pulse update live and see how they changed with simple actions like light exercise or holding our breath, for example the increase in heart rate after stepping in place was clearly visible on the plot. The cloud dashboard also stayed in sync during testing on our local WiFi network, showing that the device can stream data to a web interface for basic remote monitoring.

V. Conclusion

The goal of this project was to build a portable, Arduino based single lead ECG and heart rate monitoring prototype.

The device integrates a dedicated analog front end from our AD8232 for ECG signal acquisition with an optical heart rate and SpO_2 sensor from our MAX30102 sensor, demonstrating the ability to capture both electrical and optical cardiac signals at the same time. Our project achieved real time visualization of the ECG waveform and BPM, both locally on a serial plot and remotely via an IoT dashboard which can be viewed from either a computer or mobile device such as a smartphone.

The integration of these sensors on the Arduino platform highlights the feasibility of low cost monitoring solutions for the healthcare industry. We showed that even with simple filtering and code, the essential features of an ECG can be recorded and a reasonably accurate heart rate can be obtained. We built the prototype with about $50 in parts, which shows that this kind of system could become a low cost home monitoring tool in the future, once all current bugs are fixed and we improve the sensor accuracy.

There are several improvements we still want to try. First, adding digital band pass and notch filters that would help cut down on any noise and interference which will give us much cleaner ECG and PPG waveforms. Also implementing a more advanced peak detection algorithm such as the Pan-Tompkins algorithm for QRS detection could improve heart rate accuracy and reliability. Secondly, the user interface can be enhanced by adding a display on the device to show the heart rate and signal status in real time, eliminating the need for additional devices. Third, the wireless capabilities of the Arduino's WiFi can be fully used so that the system can stream data to a smartphone app or cloud in real time over WiFi or Bluetooth, enabling continuous remote monitoring and alerts. An improved IoT dashboard as shown in Fig. 6 could allow users to observe the ECG and heart rate data from anywhere. Additionally, data logging functionality could be added to record long-term heart data for offline analysis by adding a sd card.

While the prototype is not a replacement for medical devices, it demonstrates the core functionality of a personal heart monitor and lays the groundwork for more advanced developments in health monitoring technology. With further refinement, such Arduino based systems could become useful for fitness tracking, preliminary health screening, and much more

REFERENCES

[1] World Health Organization, "Cardiovascular diseases (CVDs) Fact Sheet," Jul. 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

[2] Analog Devices, *AD8232: Single-Lead Heart Rate Monitor Front End* (product datasheet), Rev. D, 2015.

[3] Maxim Integrated (Analog Devices), *MAX30102: High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health*, Datasheet Rev. 1, Nov. 2018.

[4] E. A. Ashley and J. Niebauer, *Cardiology Explained*. London: Remedica, 2004, ch. 3, "Conquering the ECG."

[5] A. Singh, D. Dwivedi, C. Shweta, and A. Sharma, "IoT based wireless ECG monitoring system," *Journal of Emerging Technologies and Innovative Research*, vol. 8, no. 6, pp. 904–909, 2021.

[6] Instructables, "ECG Monitoring System by Using Arduino or AD8232," [Online]. Available: https://www.instructables.com/ECG-Monitoring-System-by-Using-Arduino-or-AD8232/.